



9/1/2026

N + - Network Fundamentals & Building Networks.

Network Troubleshooting Management
Monitoring and Optimization.

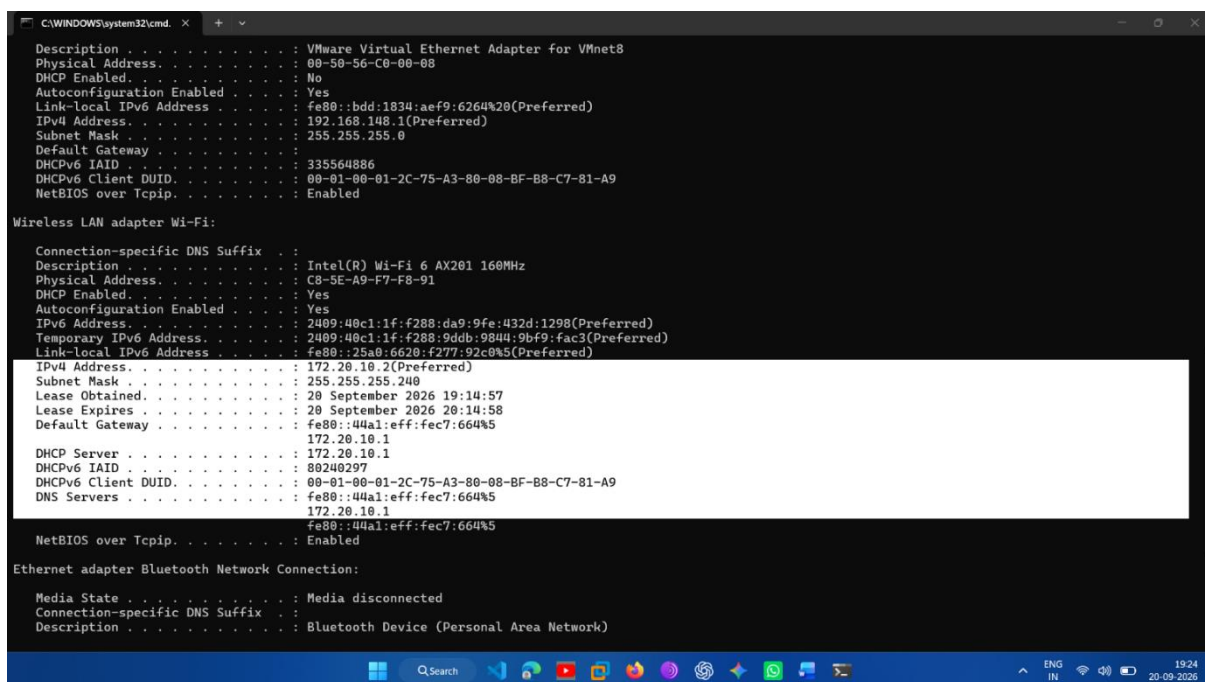
TASK 1.

- ❖ Use the Windows command prompt to run the command `ipconfig /all` on your computer and take a screenshot of the output. Identify and write down your IPv4 address, subnet mask, default gateway, and DNS servers as shown in the result.

➤ IP Configuration Using `ipconfig /all`:

Command Used:

`ipconfig /all`



```
C:\WINDOWS\system32\cmd. x + v
Description . . . . . : VMware Virtual Ethernet Adapter for VMnet8
Physical Address. . . . . : 00-50-56-C0-00-08
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::bdd:1834:aef9:6264%20(Preferred)
IPv4 Address. . . . . : 192.168.148.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 335564886
DHCPv6 Client DUID. . . . . : 00-01-00-01-2C-75-A3-80-08-BF-B8-C7-81-A9
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . . : 
Description . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
Physical Address. . . . . : C8-5E-A9-F7-F8-91
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . . : Yes
IPv6 Address. . . . . : 2409:40c1:1f:f288:da9:9fe:432d:1298(Preferred)
Temporary IPv6 Address. . . . . : 2409:40c1:1f:f288:9ddb:9844:9bf9:fac3(Preferred)
Link-local IPv6 Address . . . . . : fe80::25a0:6620:f277:02c0%5(Preferred)
IPv4 Address. . . . . : 172.20.10.2(Preferred)
Subnet Mask . . . . . : 255.255.255.240
Lease Obtained. . . . . : 20 September 2026 19:14:57
Lease Expires . . . . . : 20 September 2026 20:14:58
Default Gateway . . . . . : fe80::44a1:eff:fec7:664%5
                          172.20.10.1
DHCP Server . . . . . : 172.20.10.1
DHCPv6 IAID . . . . . : 80240297
DHCPv6 Client DUID. . . . . : 00-01-00-01-2C-75-A3-80-08-BF-B8-C7-81-A9
DNS Servers . . . . . : fe80::44a1:eff:fec7:664%5
                          172.20.10.1
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . . : 
Description . . . . . : Bluetooth Device (Personal Area Network)
```

Figure : `ipconfig /all` Output.

Result:

- ⌘ IPv4 Address: 172.20.10.2
- ⌘ Subnet Mask: 255.255.255.240

⌘ **Default Gateway:** 172.20.10.1

⌘ **DNS Server:** 172.20.10.1

TASK 2.

❖ **Simulate a network connectivity issue by disconnecting your Wi-Fi or Ethernet cable, then use the ping command to check connectivity to www.google.com. Reconnect and repeat the ping. Write a short note on the difference in results and what each status means.**

➤ **Simulate Network Connectivity Issue:**

Wi-Fi Disconnected

Command:

ping www.google.com

Result:

Request timed out.

Meaning: The computer could not reach Google because the network connection was disconnected.

Conclusion: The website with the lower response time responds faster.

Wi-Fi Reconnected

Command:

ping www.google.com

Result:

Reply from ...

Meaning: The connection was restored, and the computer could successfully communicate with Google.

Difference:

- ✓ When the network was **disconnected**, the ping failed. After reconnecting, the ping received **successful replies**, confirming that the internet connection was working again.

TASK 3.

❖ **Open Task Manager or Resource Monitor on your system and monitor network usage for 5 minutes while streaming a YouTube video. Record the peak network usage value and note which process used the most bandwidth during this time.**

➤ **Monitor Network Usage:**

Result:

Item	Observation
Monitoring Time	5 minutes
Peak Network Usage	10 Mbps
Process Using Most Bandwidth	Google Chrome

Conclusion: While streaming YouTube, Google Chrome used the most network bandwidth, with a peak usage of approximately 10 Mbps.

TASK 4.

❖ **Download and install Wireshark (or use an online packet capture demo if installation is not possible). Capture at least 30 seconds of network traffic while browsing Flipkart or Instagram Web, then filter the packets to show only HTTP or HTTPS traffic. List 3 types of information you can see in these filtered packets, Use the filter bar in Wireshark and try http or https as the filter keyword.**

➤ **Wireshark Packet Filtering:**

Filter Used

http or https

Information Observed

- ⌘ **Source and Destination IP addresses** – Shows where the packet came from and where it is going.
- ⌘ **Port Numbers** – Shows the communication ports used by the connection.
- ⌘ **Packet Size and Protocol Details** – Shows the size of the packet and information about the HTTP/HTTPS protocol.

Result: The traffic was captured for at least **30 seconds** and filtered to display only **HTTP/HTTPS packets**.

TASK 5.

❖ Imagine your friend is complaining about slow internet while gaming online. Use ChatGPT or any AI tool to generate a troubleshooting checklist (at least 5 points) for optimizing home network speed and reliability. Paste the AI's response and highlight one suggestion you would try first.

➤ Home Network Troubleshooting Checklist:

AI-Generated Checklist

- ⌘ Restart the Wi-Fi router to refresh the network connection.
- ⌘ Use a wired Ethernet connection for gaming to reduce lag and interference.
- ⌘ Move closer to the router or place the router in a better location.
- ⌘ Stop background downloads and streaming on other devices.
- ⌘ Update the router firmware and network drivers to improve stability.

First suggestion I would try: Use a **wired Ethernet connection** because it provides a more stable connection for gaming.